

Cybersecurity

Introduction to Cryptography

Lesson 11.1



Learning Objectives

By the end of today's lesson, you will be able to:

- 1 Use basic transcription and substitution ciphers and keys to encrypt simple messages.
- 2 Understand how encryption supports secure communication through the PAIN framework.
- 3 Differentiate between encoding and encrypting.
- 4 Calculate the strength and efficiency of various encryption levels.
- 5 Use the symmetric encryption tool OpenSSL to confidentially transmit secure messages.



Welcome



Introduction to **Cryptography**

Always Remember the CIA!



Confidentiality is focused on keeping information and communication secure from unauthorized parties.

The Importance of Confidentiality

It is critical for organizations to keep private information secure.

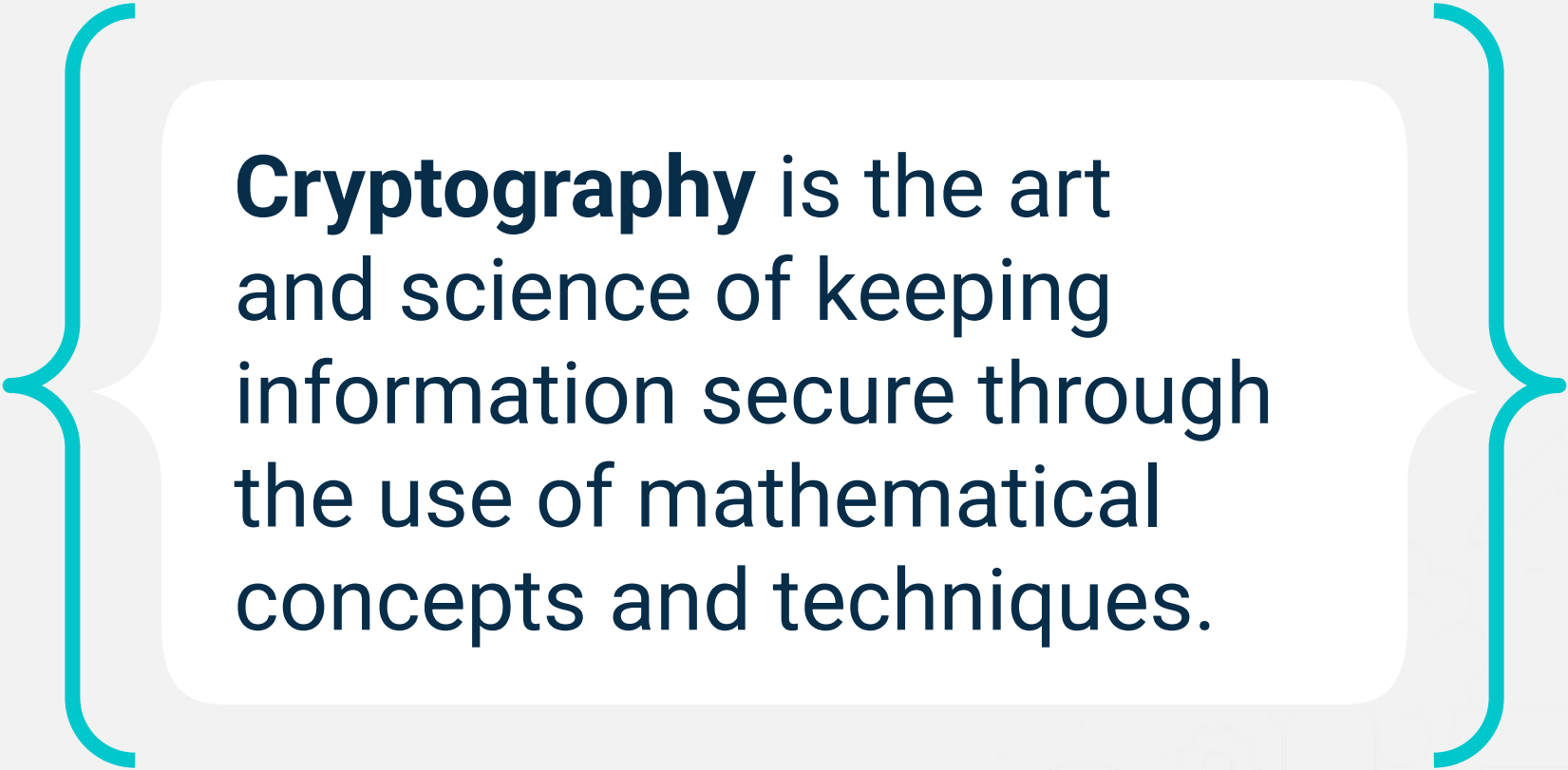
An example:

- 1 A doctor loses a laptop containing private medical records.
- 2 An unauthorized person finds the laptop, opens it, and is able to access the private data.
- 3 This can negatively impact the reputation of the doctor and the hospital responsible for the data.
- 4 This can also have a financial impact, as a leak can result in significant legal fines for disclosure of sensitive data.





A primary method for
keeping information
secure is **cryptography**.



Cryptography is the art and science of keeping information secure through the use of mathematical concepts and techniques.



This Week's Scenario

You will be playing the role of a security analyst at the Hill Valley Police Department.

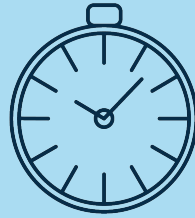
- You will be investigating—and hopefully apprehending—the **Alphabet Bandit**, who is likely responsible for a number of burglaries in Hill Valley.
- The Alphabet Bandit likes to leave hidden messages after each burglary, and we must use cryptographic techniques to investigate the incidents.





Questions?





The **History** of Cryptography



A primary method for one of the earliest methods of applied cryptography is the **Caesar cipher**.



Caesar Cipher

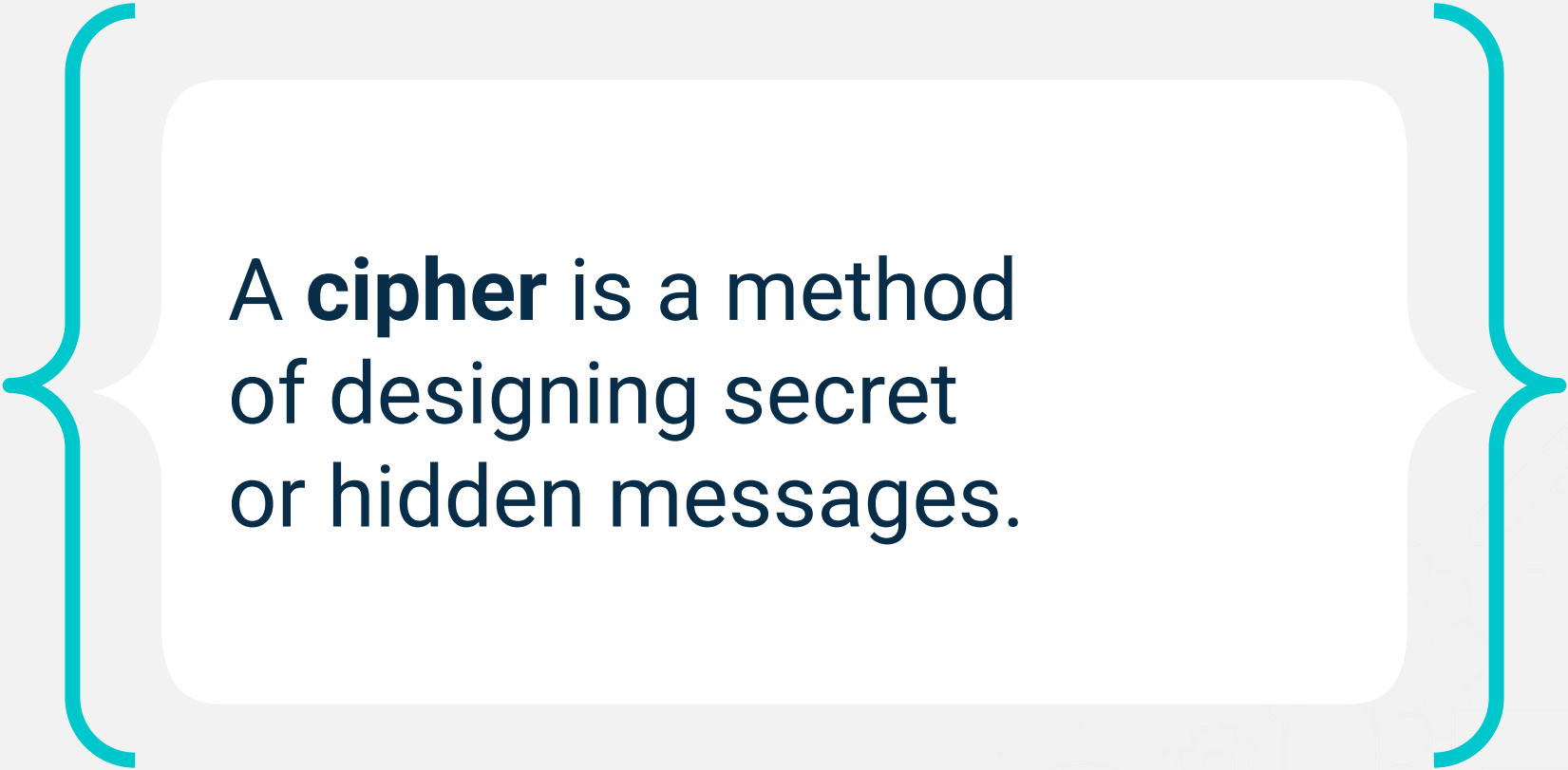
Julius Caesar needed a method to **securely** communicate with his military.

Suppose Caesar needed to tell his military to "Launch an attack at sunrise."

Caesar's plaintext:

"Launch an attack at sunrise."

In an effort to communicate with his military, the Roman general developed a **cipher** to hide his communication.



A **cipher** is a method
of designing secret
or hidden messages.





Caesar Cipher

Caesar used encryption to ensure the plaintext message was unreadable.

Caesar's plaintext:

"Launch an attack at sunrise."



Encryption

Encrypted ciphertext:

"Odxqfk dq dwxdfn dw vxqulvh."

Even if Caesar's enemies intercept the ciphertext, they would not understand the message.



Encryption is a process of modifying a message or information to prevent unauthorized parties from accessing it.



The goal of the Caesar cipher is not only to prevent unauthorized parties from reading the communication, **but to also allow** authorized parties to receive and understand the hidden message.



Caesar Cipher

Caesar's military would receive the ciphertext and convert back to plaintext.

Encrypted ciphertext:

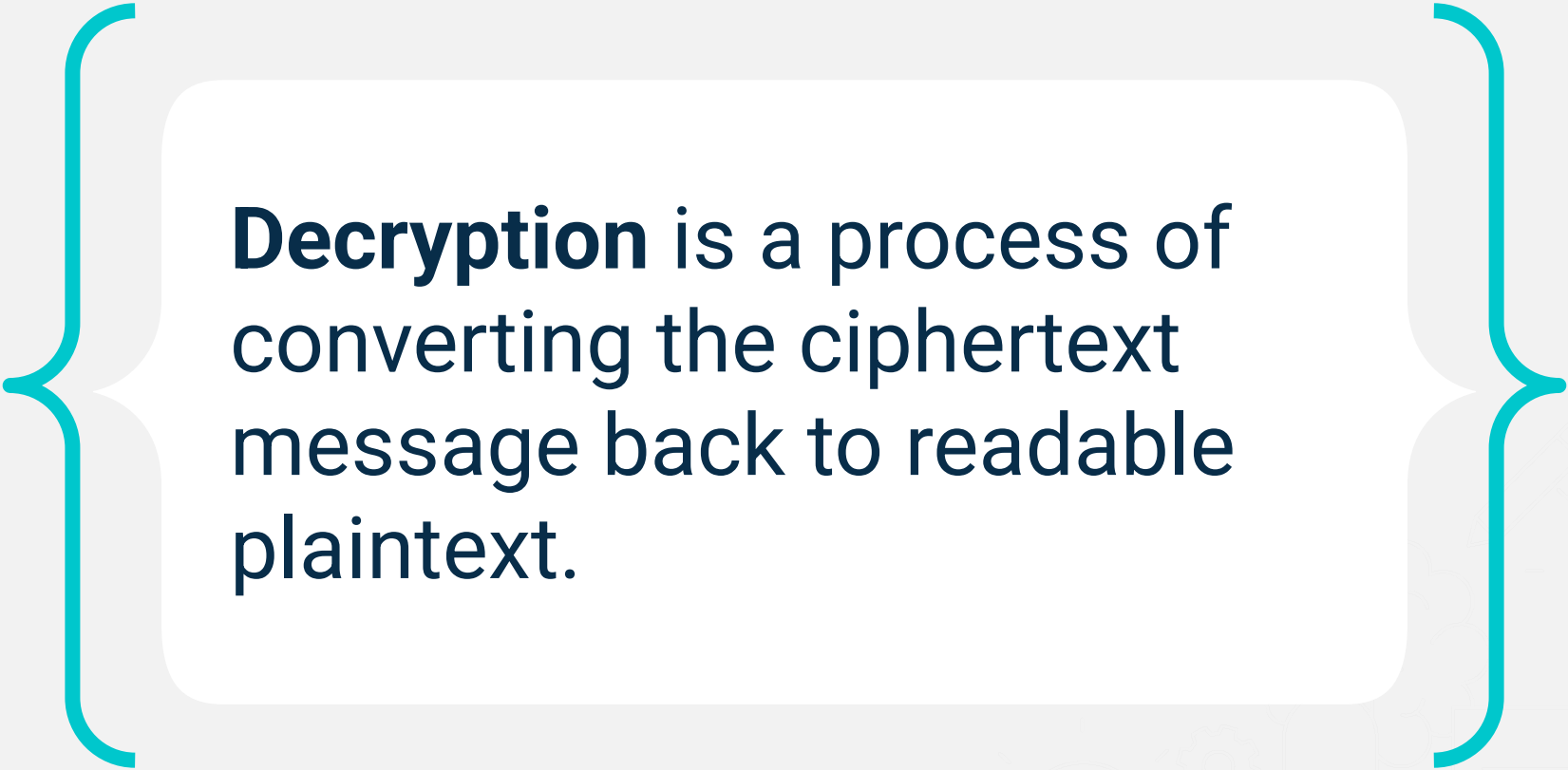
"Odxqfk dq dwwdfn dw vxqulvh."



Decryption

Caesar's plaintext

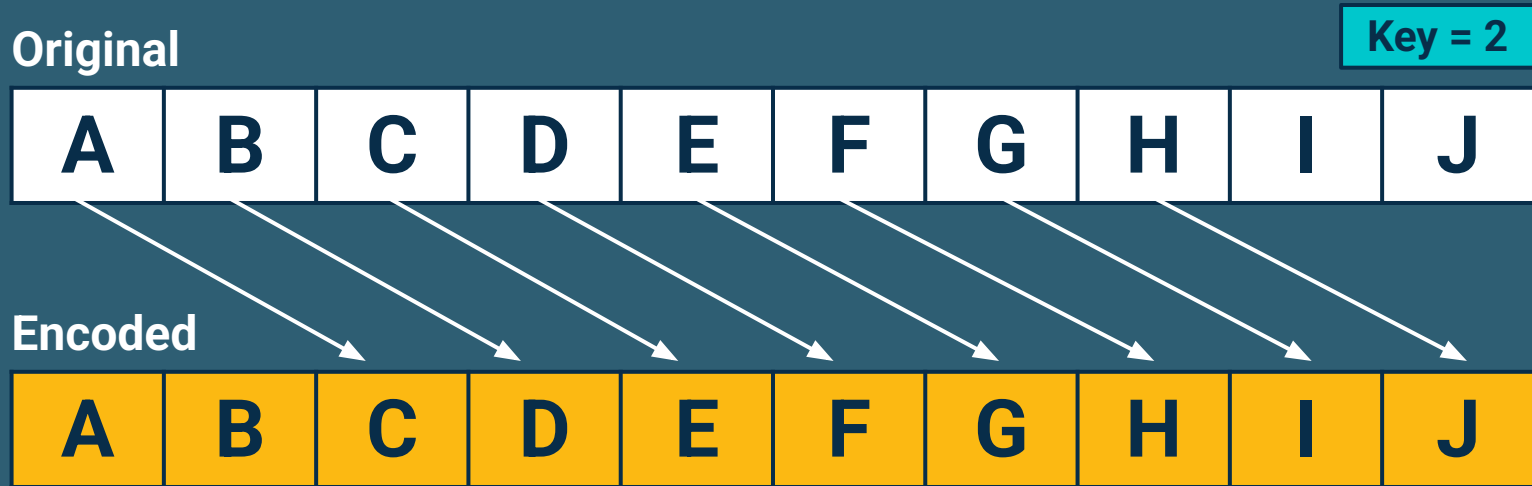
"Launch an attack at sunrise."



Decryption is a process of converting the ciphertext message back to readable plaintext.

How the Caesar Cipher Works

The Caesar cipher works by shifting letters a set number of positions (**the key**) from the original letter.



Examples

"I HID A BAG" → "K JKF C DCI"

"A BAD DAD" → "C DCF FCF"

Summary: Key Cryptographic Terms

Plaintext	Information in a human-readable form.
Ciphertext	A plaintext message that has been encrypted into an unreadable form.
Encryption	The process of converting plaintext to ciphertext.
Decryption	The process of converting ciphertext to plaintext.
Cipher	A method of performing encryption or decryption.
Key	A parameter specifying how plaintext is converted to ciphertext and vice versa.
Caesar cipher	A type of cipher that shifts the letters in the alphabet by a fixed number.



Questions?





Activity

Caesar cipher code names

In this activity, you will play the role of a security analyst working for the Hill Valley Police Department.

You have been assigned to a top secret task force to find the Alphabet Bandit. You must create a code name, encrypt it, and send it to your partner.

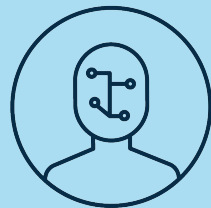
Suggested Time
10 minutes



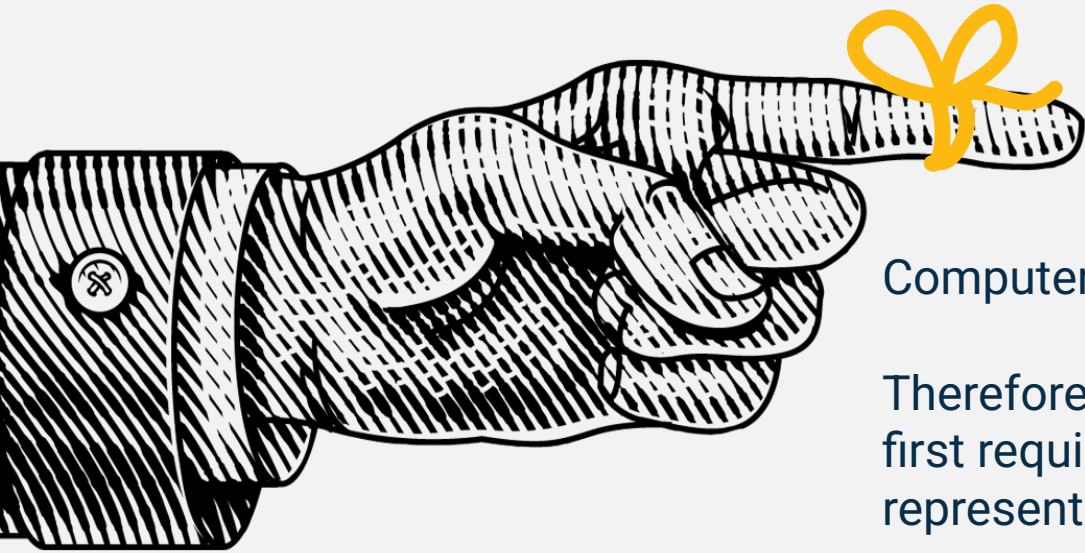


Questions?





Introduction to **Character** Encoding



Remember

Computers transmit digital data with **binary**.

Therefore, encrypting data on computers first requires a method of alphanumeric representation known as **character encoding**.



Character Encoding

While encoding may seem similar to encryption, they have very different goals:

Encoding

- Used to transform data so that it can be properly used by different types of systems
- Not used to keep information secret
- Data is encoded with publicly available schemes that can be decoded by anyone

Does not use a key.



Encryption

- Used to keep information from being accessed by unauthorized parties

Uses a **key** to encrypt and decrypt.





Character Encoding

There are many encoding schemes available:

Binary

ASCII

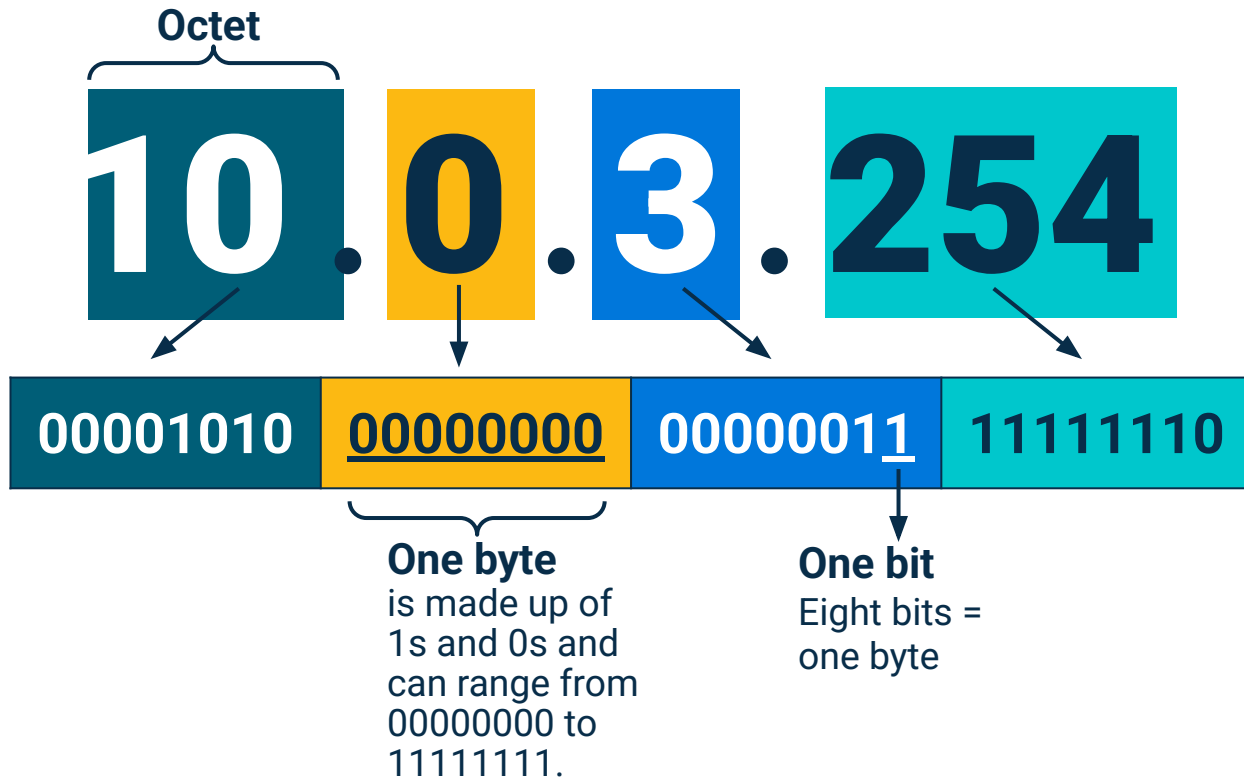
Hex

Octal



Binary Encoding

Binary is the basis of digital communications





Binary Encoding

This conversion is called binary to decimal encoding

The byte

00000000

represents the decimal **0**.

The byte

11100000

represents the decimal **224**.

The byte

11111111

represents the decimal **255**.

Since each byte can only represent a number from zero to 255, and because we read text with letters, we use **ASCII encoding** to convert these numbers to letters.

What is ASCII Encoding?

ASCII encoding is the representation of every upper and lowercase letter of the English alphabet, as well as common punctuation marks and graphic and mathematical symbols, as a number between zero and 255.

Note: 0–127 is considered standard ASCII and 127–255 is considered extended ASCII.



ASCII AMERICAN STANDARD CODE FOR INFORMATION INTERCHANGE

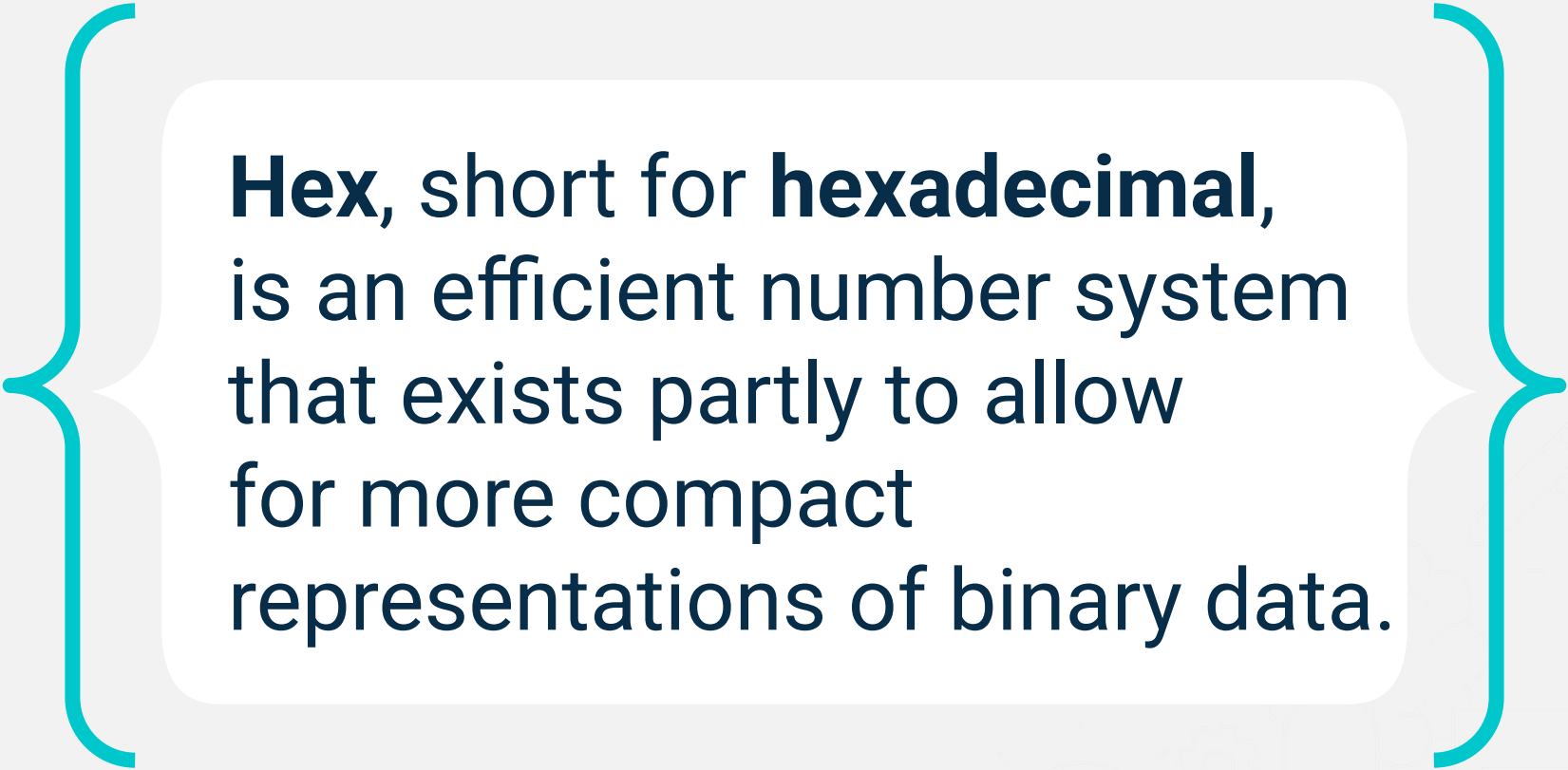


ASCII Encoding

ASCII represents computer-stored characters in a human-readable format

Look down at your keyboard. Every character is part of ASCII.
For example, upper and lowercase letters (AaBbCc...),
special characters (!@#\$...), and numbers (1,2,3,4...).





Hex, short for **hexadecimal**,
is an efficient number system
that exists partly to allow
for more compact
representations of binary data.



Hex Encoding

The hex system uses **16 symbols** to represent the base values

0 1 2 3 4 5 6 7 8 9

The **base numbers**
range from 0–9.

A B C D E F

The **letters A–F**
represent 10–15.

For example:

0x52 = R

and

0x7A = z



Hex Encoding

Hex representation of binary data

Binary representation has the disadvantage of very long numbers:

1101000 1100101 1101100 1101100 1101111

hello

hello = 68 65 6c 6c 6f

This conversion is called **hex-to-ASCII** encoding or **hex-to-text** encoding.



Octal Encoding

The octal system applies a principle similar to hex, but uses digits 0 through 7

Octal	=	Decimal
042		34



Additional Character Encoding Techniques

There are other character encoding techniques you may encounter:

Base64

- Encodes binary data to ASCII characters
- Used for storing or sending binary data

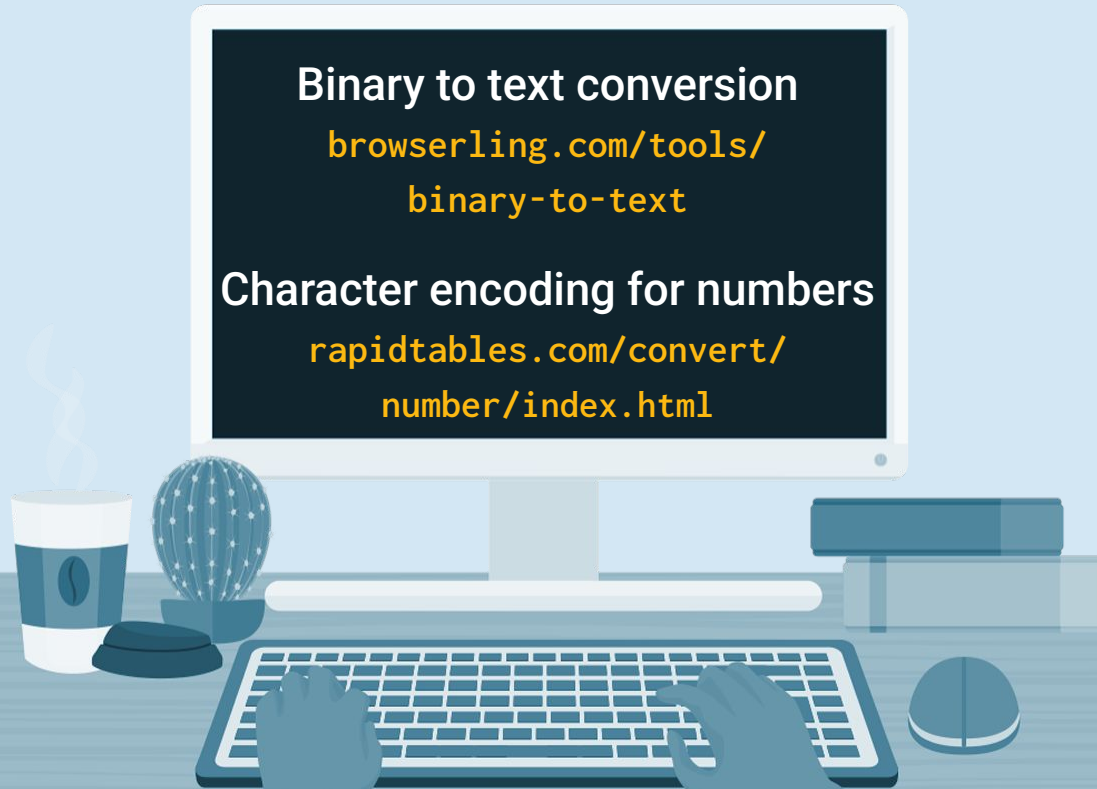
URL Encoding

- Modifies characters so they can be transmitted over the internet



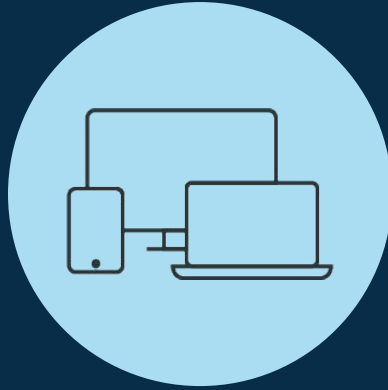
Encoding and Decoding Tools

There are various online tools to help with the encoding and decoding process





Let's practice decoding
a binary message using
browserling.com.



Instructor Demonstration

Encoding and decoding tools

Summary of Cryptographic Concepts

- The goal of **encoding** isn't to keep a message secret, but to transform data to be used by another system.
- Encoding, unlike **encryption**, does not use a key.
- Encoding is often used to transform digital text data into binary data, where encryption commonly takes place.
- There are many types of **encoding schemes** available and each is relevant for different circumstances.
- There are many free online resources for encoding and decoding messages, such as [Browserling](#) and [Rapid Tables](#).



Questions?





Activity

Decoding

You will continue in the role of a security analyst working for the Hill Valley Police Department.

- There was another burglary last night, which occurred at Mayor Wilson's residence.
- The bandit left behind an encoded message at the mayor's house.
- You are tasked with decoding the message to determine the bandit's next target.

Suggested Time
15 minutes

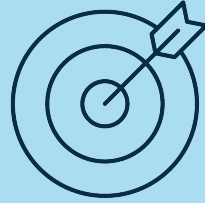


Time's Up!
Let's review



Questions?





Goals of Cryptography

Goals of Cryptography

Introducing the PAIN model



Privacy
(Confidentiality)

Authentication

Integrity

Non-repudiation

Goals of Cryptography

Introducing the PAIN model



Privacy
(Confidentiality)

Authentication

Integrity

Non-repudiation

Privacy keeps data secure from unauthorized parties.

Privacy/Confidentiality

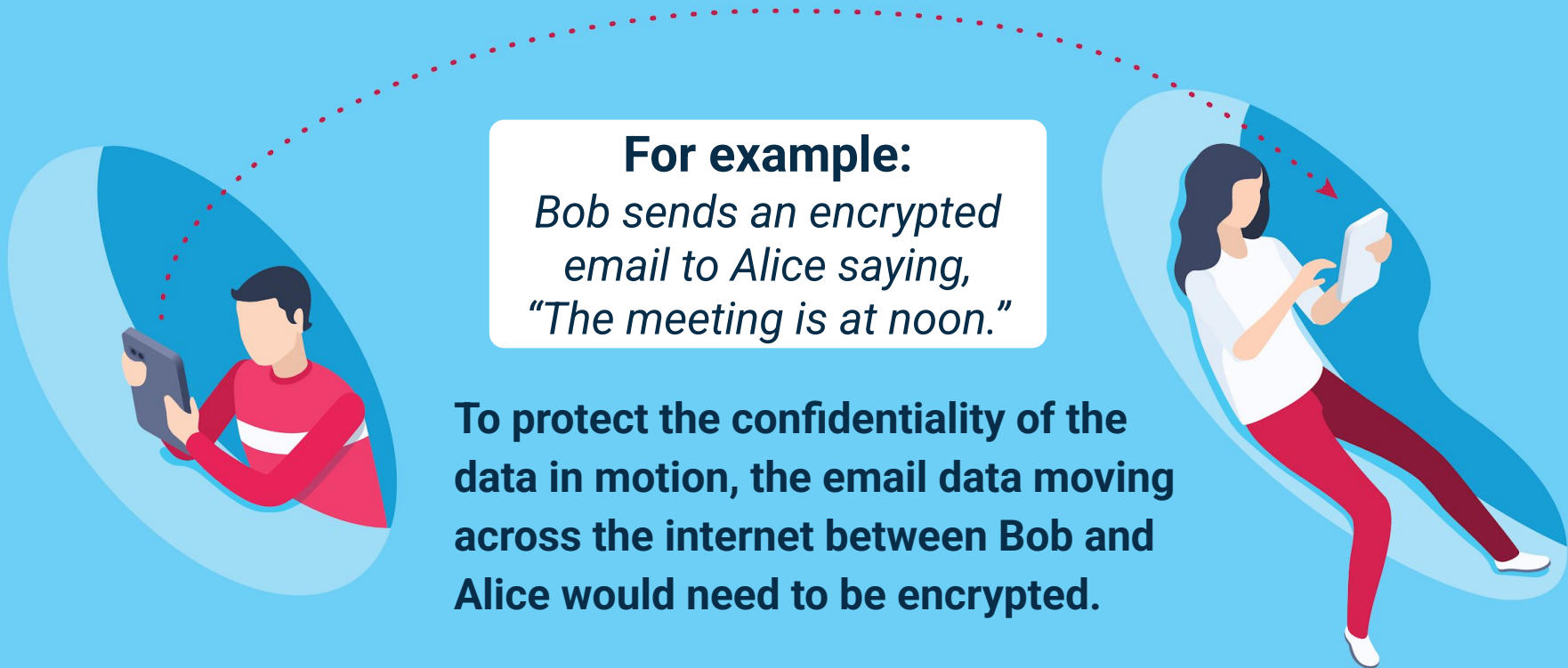
Data in motion: data moving between devices



For example:

Bob sends an encrypted email to Alice saying, "The meeting is at noon."

To protect the confidentiality of the data in motion, the email data moving across the internet between Bob and Alice would need to be encrypted.



Privacy / Confidentiality

Data at rest: static data such as that stored on a hard drive or in a database



For example:

Data stored on your laptop.

To protect the confidentiality of this data at rest, the laptop's hard drive would be encrypted.



Goals of Cryptography

Introducing the PAIN model



Authentication is used to confirm the identities of the sender and receiver of data.

Authentication

An attacker can still send encrypted data and claim they are someone they are not

PAIN

01

Bob sends a message to Alice:
"The meeting is at noon."

02

Scammer Tim sends an email impersonating Bob:
"The meeting is cancelled."

03

Alice receives the email.

Without authentication, Alice could be tricked into thinking the meeting is cancelled.

Disruptive Hacker



Goals of Cryptography

Introducing the PAIN model



Integrity ensures a message isn't altered between when it's sent and when it's received.

Integrity

Even if a message is encrypted and the sender is authenticated, an attacker can still alter the contents of a message.



01

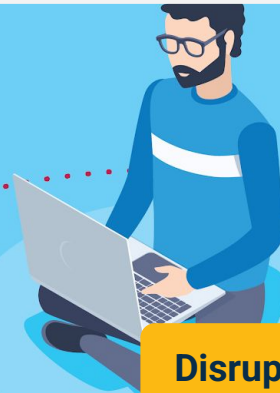
Bob sends a message to Alice: **"The meeting is at noon."**

02

Scammer Tim intercepts the message and sends an email impersonating Bob: **"The meeting is at 5am."**

03

Alice receives the email.



Goals of Cryptography

Introducing the PAIN model

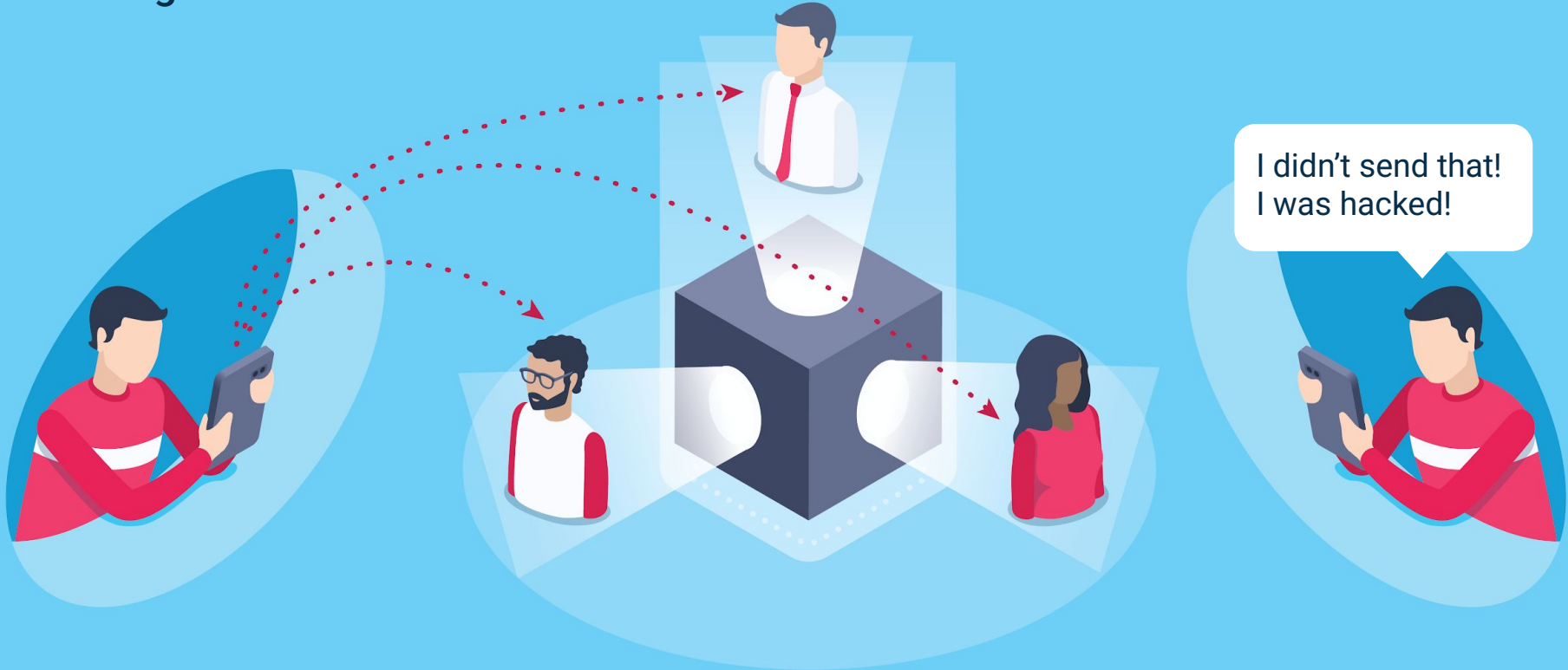


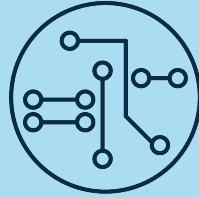
Non-repudiation prevents the original sender from denying they were the sender.

Non-Repudiation

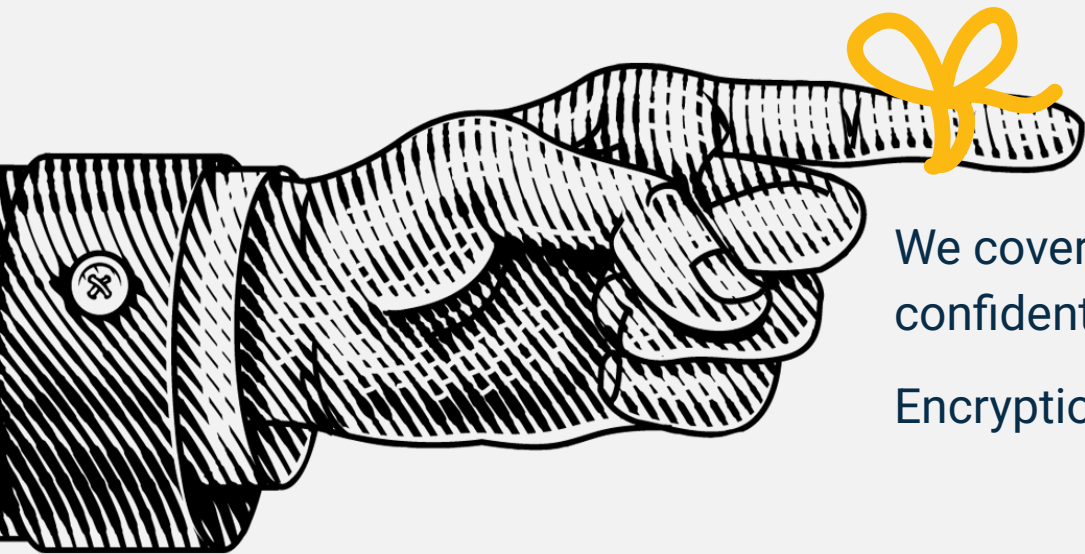
For example, an employee sends an inappropriate message to their entire office.

PAIN





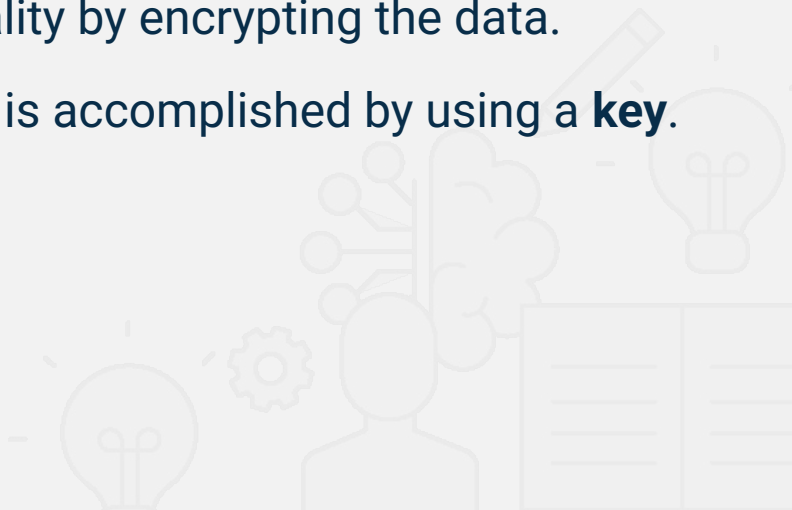
Cryptography **Ciphers**

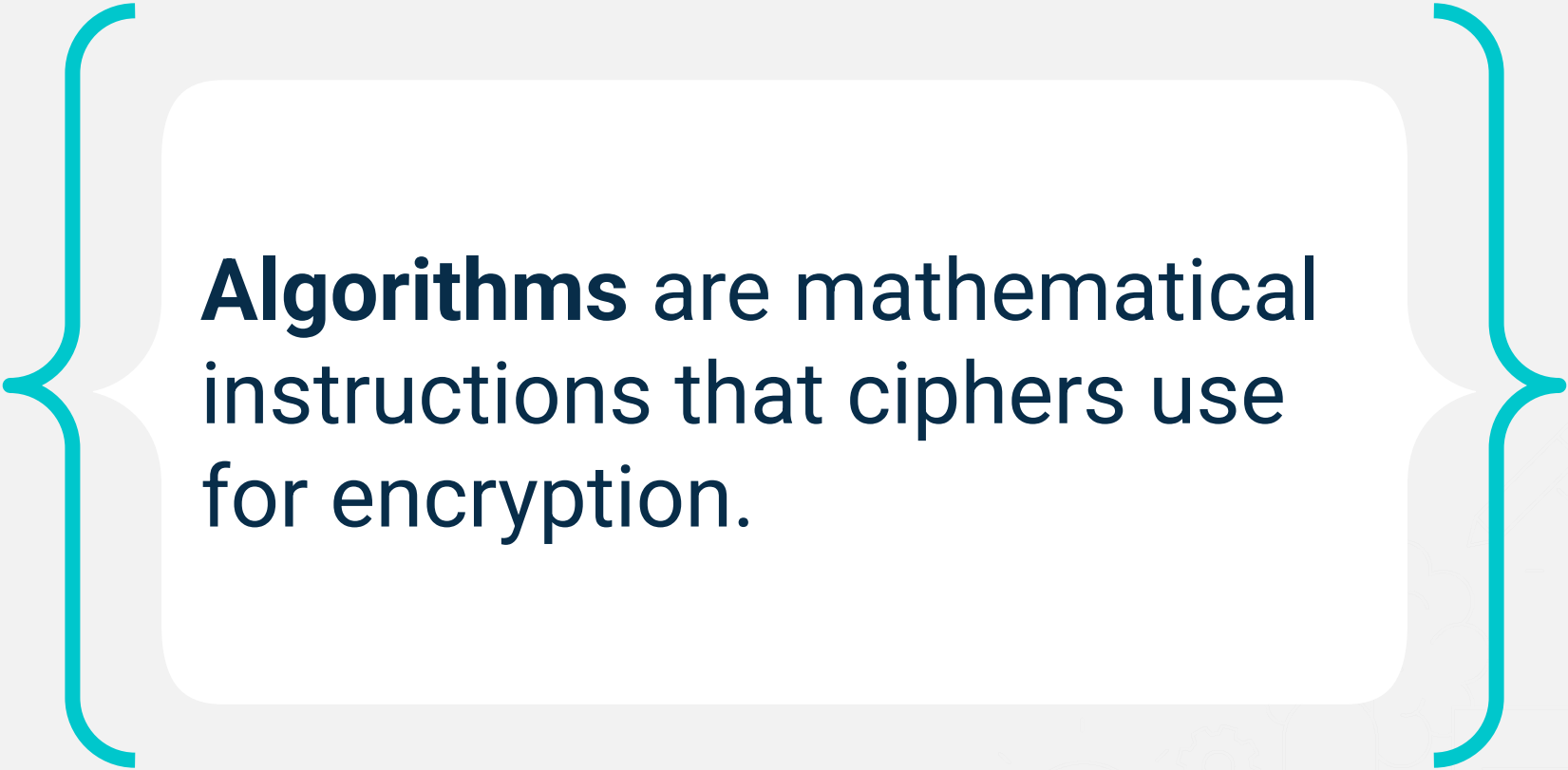


Remember

We covered how **ciphers** are used to provide confidentiality by encrypting the data.

Encryption is accomplished by using a **key**.





Algorithms are mathematical instructions that ciphers use for encryption.



Introduction to Cryptography Ciphers

Types of ciphers:

01

Stream ciphers

02

Block ciphers

03

Substitution ciphers

04

Transposition ciphers

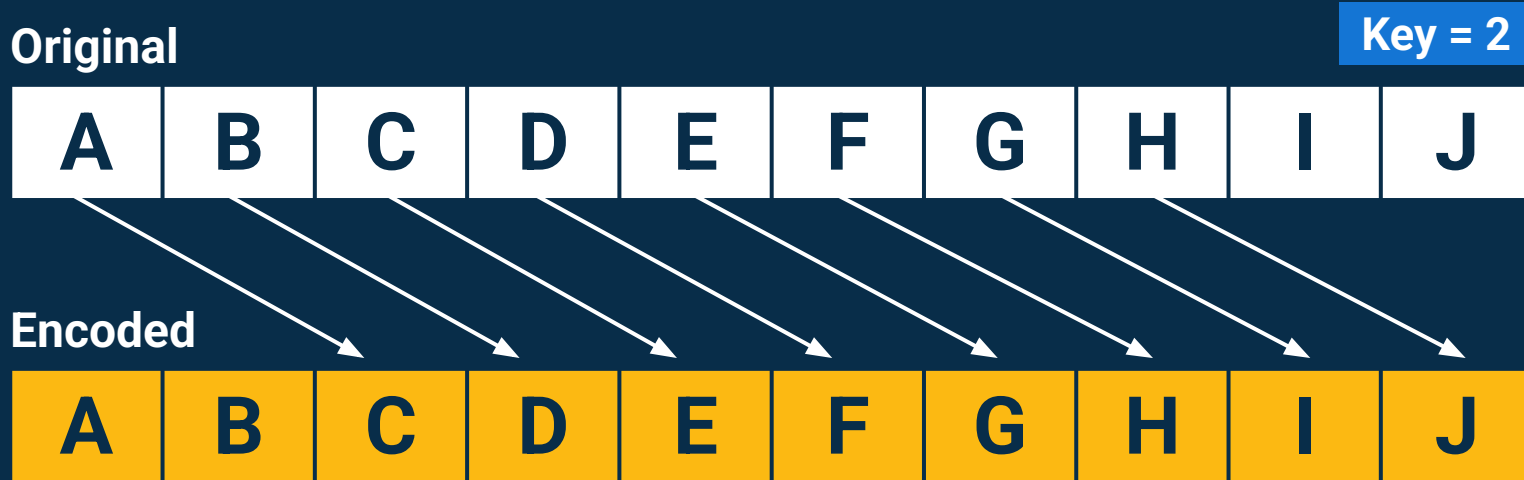


Stream ciphers apply their algorithm one bit or one character at a time.

A prominent type of stream cipher is the **substitution cipher**.

Stream and Substitution Ciphers

- Caesar cipher is a **substitution cipher**
- While substitution alone doesn't provide strong encryption, when combined with other techniques, it can result in strong, fast encryption





Block ciphers apply their algorithms to chunks, or blocks, of characters.

A prominent type of block cipher is the **transposition cipher**.

Block and Transposition Ciphers

- Break the message into blocks of three characters
- Replace the first, second, and third character of each block with the third, first, and second character respectively
- Combine the rearranged text

Key =

1	2	3
3	1	2

Message

Encrypted
Message

H	E	L	L	O	!
L	H	E	!	L	O

Summary

We're learning a lot of new concepts this week. Let's review!

01

The goals of cryptography are illustrated with the **PAIN model**.

02

PAIN is an acronym for privacy, authentication, integrity, and non-repudiation.

03

Ciphers use mathematical formulas known as **algorithms** to encrypt and decrypt data.

04

The main cipher categories are **block ciphers** and **stream ciphers**.

05

Stream ciphers apply their algorithms one character at a time, while block ciphers apply the algorithms to blocks of characters.

06

One type of stream cipher is the substitution cipher. One type of block cipher is the transposition cipher.

07

Substitution ciphers replace each character with a different character.

08

Transposition ciphers rearrange the letters within a defined block size.



Questions?





Activity

Cryptography concepts and ciphers

You will continue to play the role of a security analyst working for the Hill Valley Police Department.

- Unfortunately, you weren't able to act fast enough and Dr Brown's house was burglarized.
- Your task is to use key provided to decrypt the most recent message left by the Alphabet Bandit.

Suggested Time
15 minutes



Time's Up!
Let's review



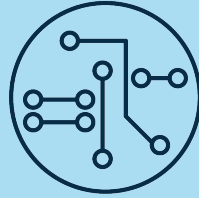
Questions?



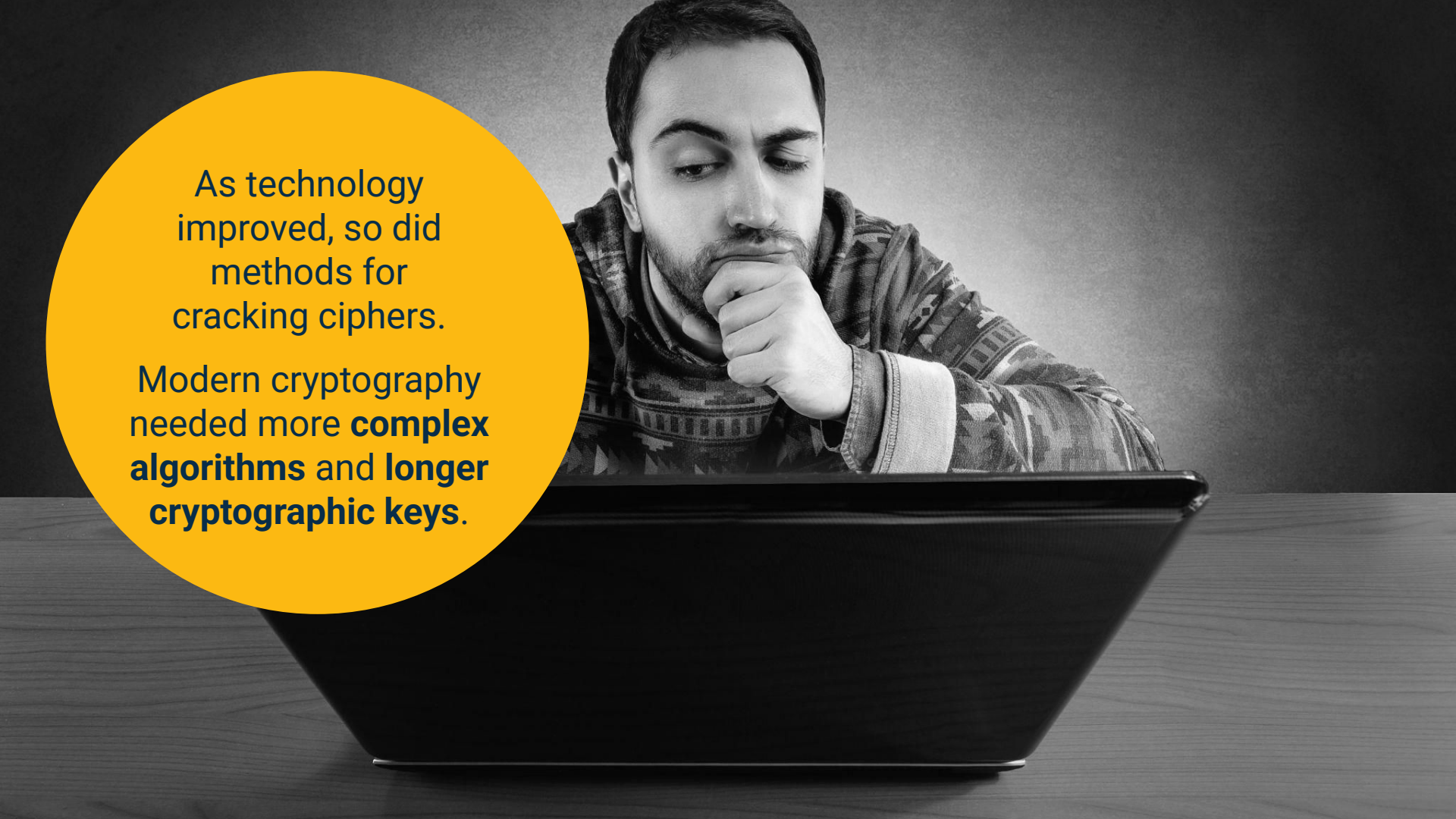


Break

15 mins

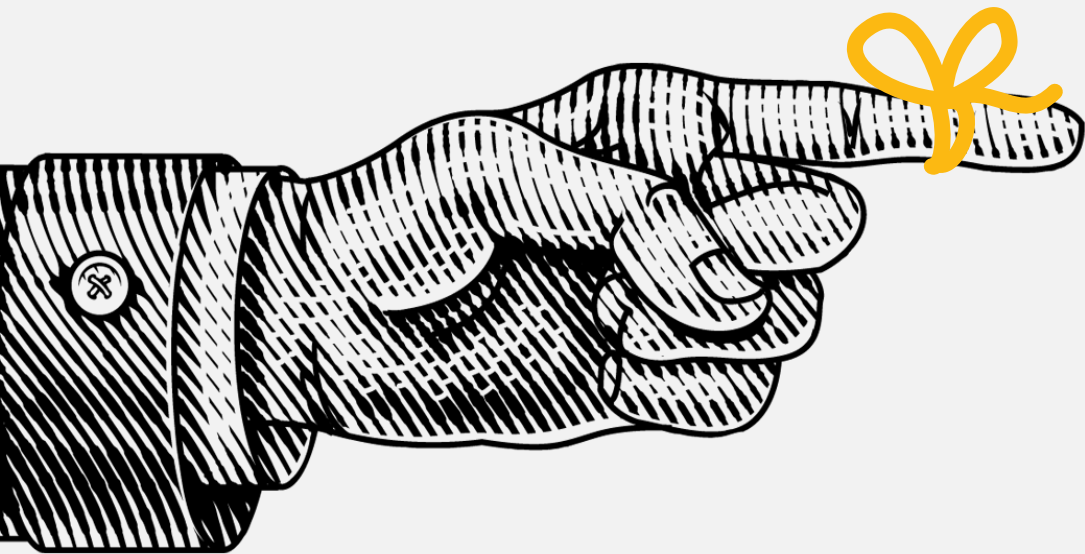


Cryptography and Cryptographic Keys

A black and white photograph of a man with a beard, wearing a patterned sweater, sitting at a desk and looking at a laptop. He has his hand on his chin, appearing to be in deep thought. The background is a dark, textured wall.

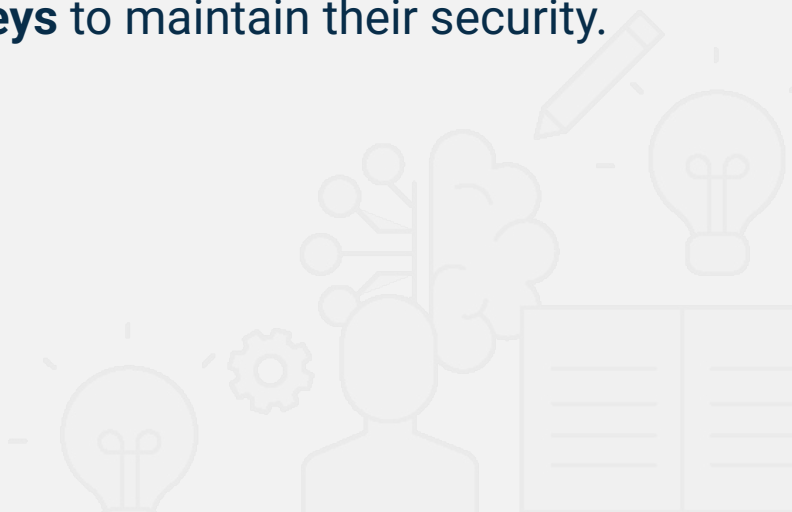
As technology improved, so did methods for cracking ciphers.

Modern cryptography needed more **complex algorithms** and **longer cryptographic keys**.



Remember

Cryptographic algorithms use **keys** to maintain their security.



Cryptographic Keys

Key space



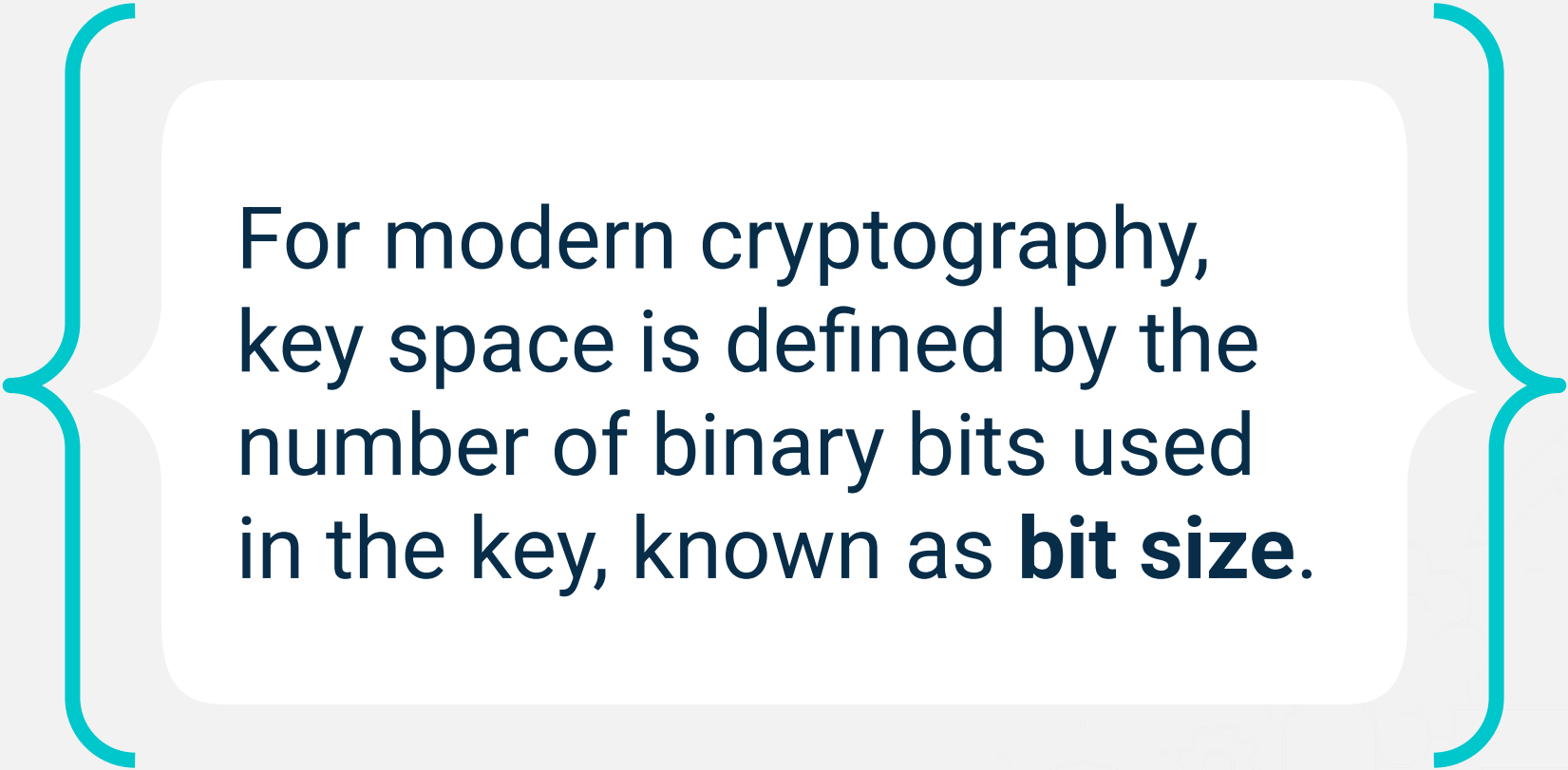
Each algorithm has a possible range of numbers that can be used as a key, known as **key space**.

For example:

If a password could only be one numerical digit, the possible values are:

[0 1 2 3 4 5 6 7 8 9]

The key space is 10.



For modern cryptography,
key space is defined by the
number of binary bits used
in the key, known as **bit size**.



A common question you may hear IT professionals ask is, "How much more secure is **x more bits** of encryption?"

Keys and Bit Size

We'll need to understand how bit size affects key space and the level of encryption

For each bit added, the key space doubles in size:

Key space = $2^{\text{bit size}}$

1 bit

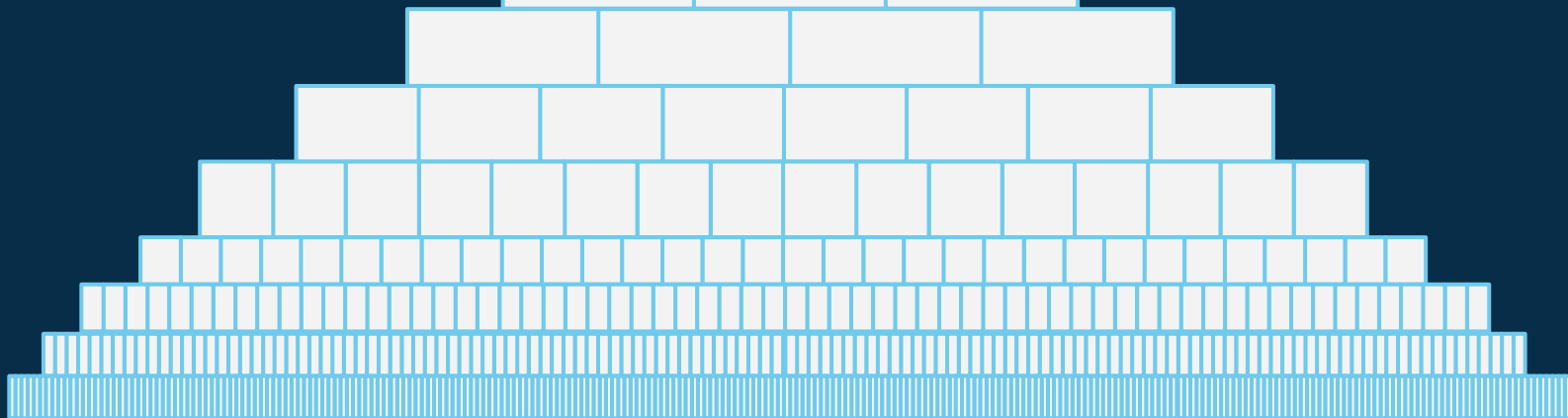
Key space = 2 (1, 0)

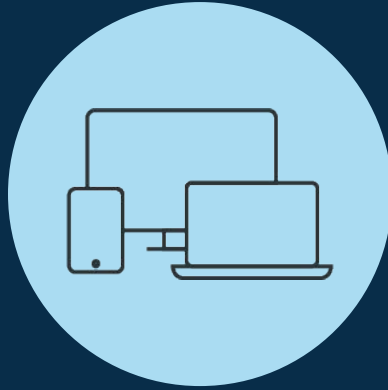
2 bit

Key space = 4 (00, 01, 10, 11)

3 bit

Key space = 8 (000, 001, 010, 100, 101, 110, 111)





Instructor Demonstration

Encryption strength



Activity

Encryption strength

You will continue to play the role of a security analyst working for the Hill Valley Police Department.

- Captain Strickland has tasked you with evaluating several email security vendors and choose the one that will best protect future emails.
- You will select a cost-efficient email security vendor and provide a summary to your police captain.

Suggested Time
10 minutes

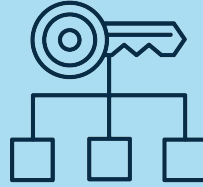


Time's Up!
Let's review



Questions?





Symmetric Key Algorithms



Why don't we just have
one million-bit key to
ensure it's never cracked?

It takes time and
computational
resources to encrypt
and decrypt larger keys.



Is the encryption
strength worth the
time it takes to use
the key?

Security Trade-Offs

Do we want an incredibly strong cipher that's hard to compute and difficult to decrypt?

Do we want "average" security that's faster?



Modern symmetric key algorithms use algorithms that are **secure** and **fast**.

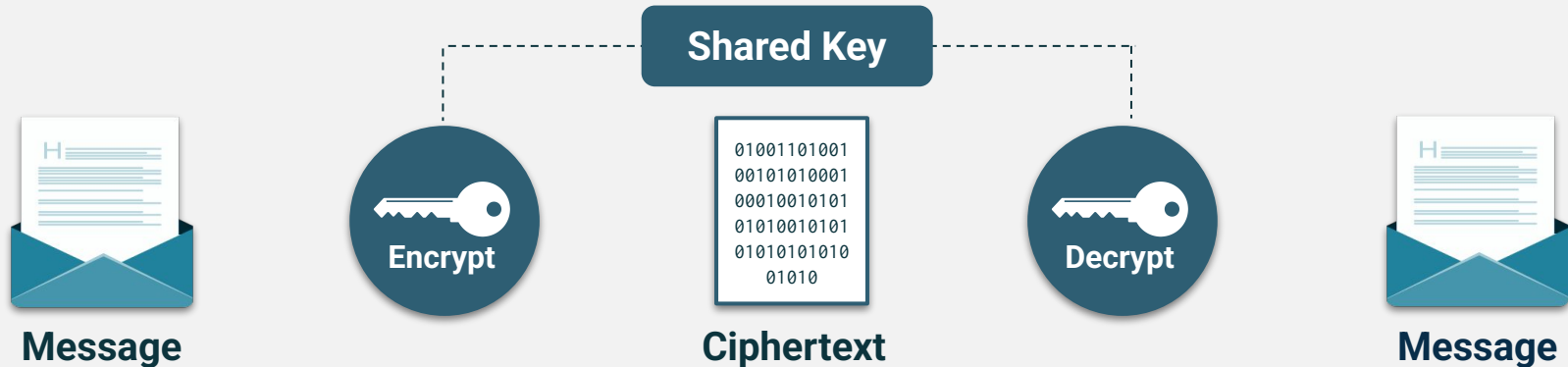


Finding the Balance

Symmetric key algorithms use a single, shared key to encrypt and decrypt a message. This shared key needs to remain **private**. If exposed, the message can be decrypted by anyone.

Some widely known symmetric key algorithms include:

- Data encryption standard (DES)
- Triple DES (3DES)
- Advanced encryption standard (AES)



Data Encryption Standard (DES)

DES is a 56-bit algorithm published by the US government in 1977.

- Flaws were found in DES and additional security was added to create Triple **3DES**.
- However, the security community banded together to develop a newer, more secure symmetric encryption algorithm.

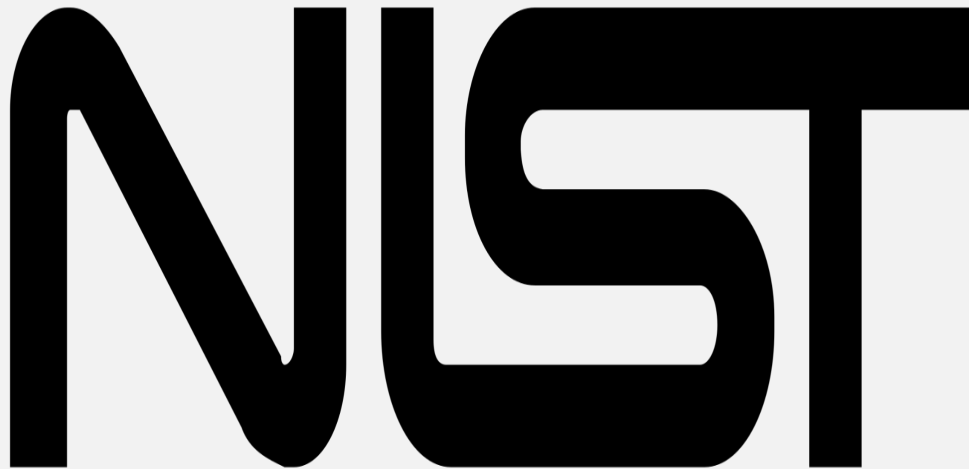


AES

Advanced Encryption Standard

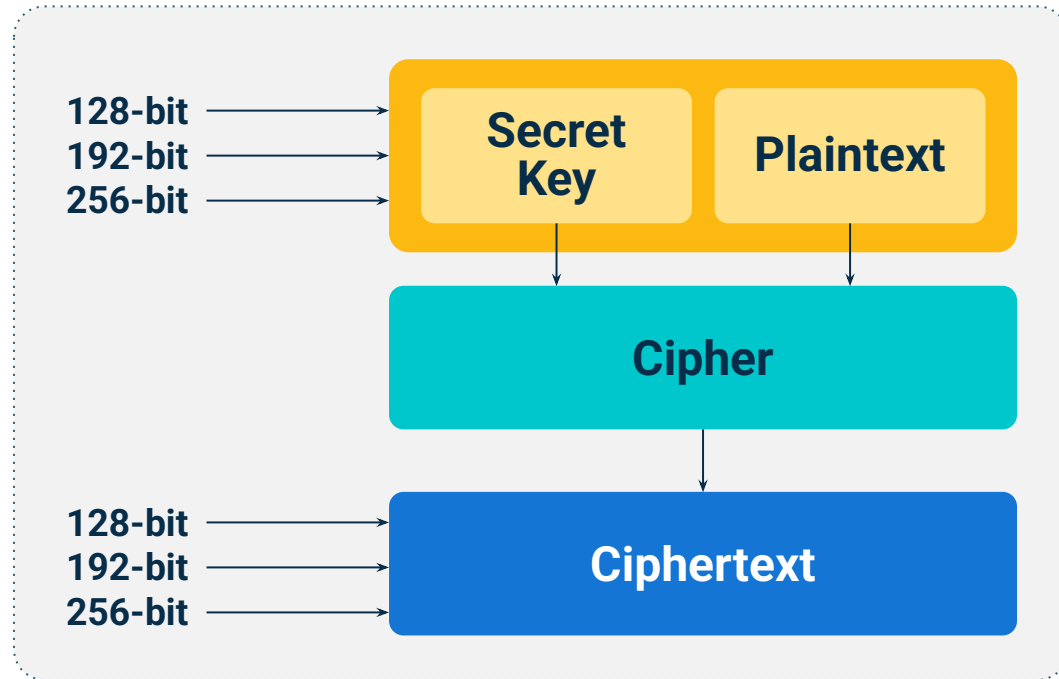
In 1997, NIST opened a contest for cryptographers to submit algorithms, with 15 collected submissions.

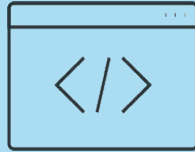
The community attempted to break them, and the five most promising were subjected to additional cryptanalysis.

The NIST logo is rendered in a large, bold, black, sans-serif typeface. The letters are thick and blocky, with a slightly rounded top on the 'N' and 'S'. The 'I' is a simple vertical bar, and the 'T' has a short horizontal crossbar. The logo is positioned on the right side of the slide, with two thin horizontal lines pointing towards it from the text blocks on the left.

Advanced Encryption Standard

Eventually, the **Rijndael cipher** was determined to be the strongest. After it was refined and standardized, it became the **advanced encryption standard (AES)**.





OpenSSL



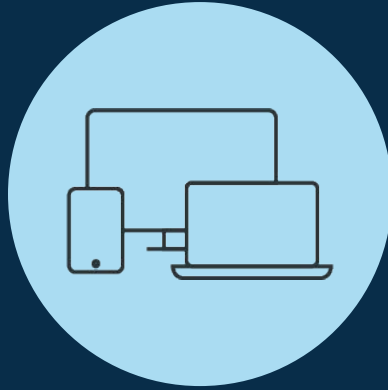
Introduction to OpenSSL

Some algorithms are used today to encrypt data

The Caesar cipher could encrypt a message by hand.

The **AES** algorithm is complicated and cannot be used without advanced technologies.

Fortunately, there is a free command line-based program called **OpenSSL** that can simplify symmetric encryption and decryption.

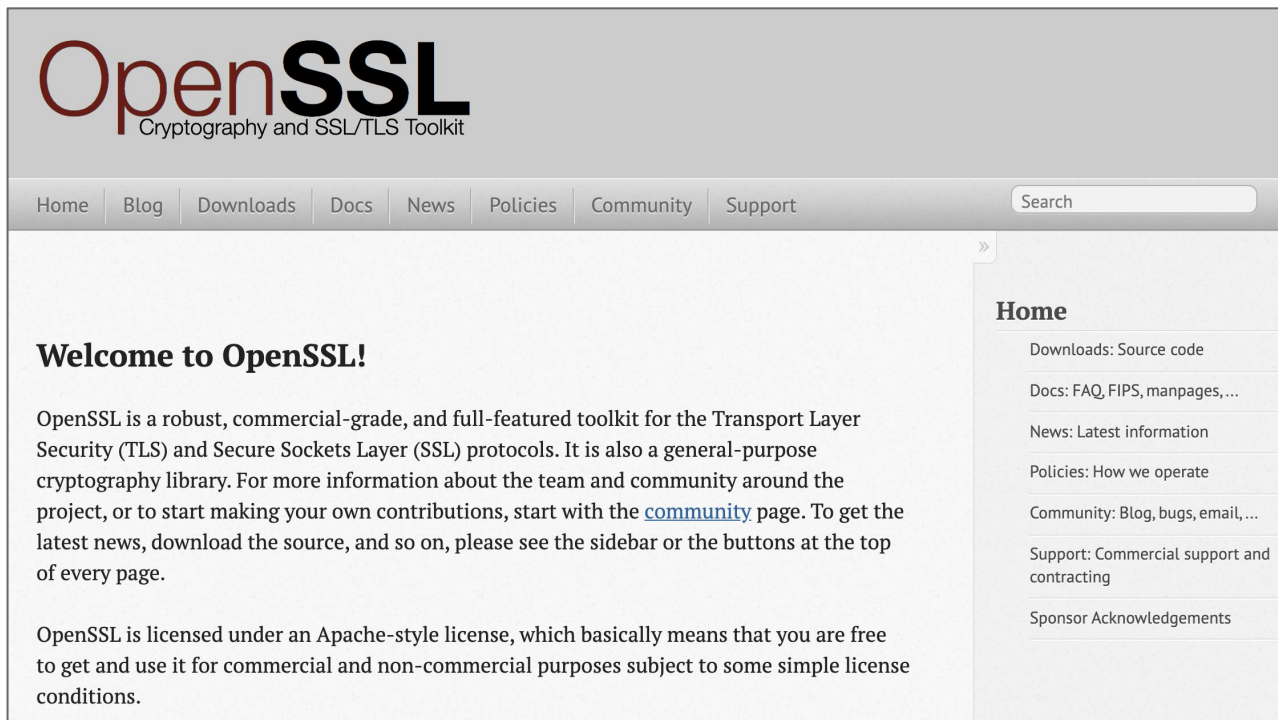


Instructor Demonstration

OpenSSL

Demo Summary

OpenSSL can generate a **random key** and **initialization vector (IV)**. With the key and IV, OpenSSL can encrypt and decrypt a message with simple terminal commands.



The screenshot shows the OpenSSL website homepage. At the top is the OpenSSL logo with the tagline "Cryptography and SSL/TLS Toolkit". Below the logo is a navigation bar with links to Home, Blog, Downloads, Docs, News, Policies, Community, and Support, along with a search bar. The main content area features a "Welcome to OpenSSL!" heading followed by a paragraph describing the toolkit. A sidebar on the right lists various resources under the heading "Home".

OpenSSL
Cryptography and SSL/TLS Toolkit

Home | Blog | Downloads | Docs | News | Policies | Community | Support

Search

Welcome to OpenSSL!

OpenSSL is a robust, commercial-grade, and full-featured toolkit for the Transport Layer Security (TLS) and Secure Sockets Layer (SSL) protocols. It is also a general-purpose cryptography library. For more information about the team and community around the project, or to start making your own contributions, start with the [community](#) page. To get the latest news, download the source, and so on, please see the sidebar or the buttons at the top of every page.

OpenSSL is licensed under an Apache-style license, which basically means that you are free to get and use it for commercial and non-commercial purposes subject to some simple license conditions.

Home

- Downloads: Source code
- Docs: FAQ, FIPS, manpages, ...
- News: Latest information
- Policies: How we operate
- Community: Blog, bugs, email, ...
- Support: Commercial support and contracting
- Sponsor Acknowledgements

Demo Summary

Creating the key and IV:

```
openssl enc -pbkdf2 -nosalt -aes-256-cbc -k mypassword -P > key_and_IV
```

Encrypting:

```
openssl enc -pbkdf2 -nosalt -aes-256-cbc -in plainmessage.txt -out plainmessage.txt.enc -base64 -K  
89E01536AC207279409D4DE1E5253E01F4A1769E696DB0D6062CA9B8F56767C8 -iv  
EE99333010B23C01E6364E035E97275C
```

Decrypting:

```
openssl enc -pbkdf2 -nosalt -aes-256-cbc -in plainmessage.txt.enc -d -base64 -K  
89E01536AC207279409D4DE1E5253E01F4A1769E696DB0D6062CA9B8F56767C8 -iv  
EE99333010B23C01E6364E035E97275C
```

Demo Summary

openssl	Initializes the OpenSSL program.
enc	Stands for “encryption.”
-pbkdf2	Specifies the encryption key type.
-nosalt	Specifies that salting will not be applied. (Salting, which we’ll cover in more depth later, adds a random value.)
-aes-256-cbc	Is the name of the cipher used.
-k PASSWORD	Creates a key with the password <code>mypassword</code> .
-P > key_and_IV	Prints out the key and IV to a file called <code>key_and_IV</code> .



Activity

Wireless attacks

In this activity, you will continue to play the role of a security analyst working for the Hill Valley Police department.

- Captain Strickland still needs to communicate with you securely, so they sent you an urgent message encrypted with OpenSSL.
- Your task is to decrypt the message and follow the instructions.

Suggested Time
20 minutes



Time's Up!
Let's review



Questions?





Next Class

Asymmetric encryption and hashing



Thank You